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Principal Component Analysis (PCA)

Motivation

Raw data are often represented in a high-dimensional space, but this representation is not always optimal for analysis or modeling.

- Basic representations may not be optimal
- Latent models allow revealing the underlying (latent) factors that govern the studied process
- This approach uses statistical correlation to identify **linear** latent factors
- This allows a simplification of data through reasoned dimensional reduction

We will explore several interpretations of this fundamental method.

Recommended reading: [1] (chapter 12) and [3] (chapters 3 and 10.3)

Formalization

Variance maximization

Geometric intuition

Given a dataset $X \subset \Omega$, we seek to decompose Ω into subspaces such that the projection of X onto these subspaces preserves the maximum amount of **information**.

Crucial question: What information should we preserve?

- Imagine that X is almost contained in a 2D plane within a 3D space
- A relevant choice would be to choose this plane as the subspace, with a basis $\{\mathbf{u}_1, \mathbf{u}_2\}$

Key characteristics of a good basis:

- The data vary **maximally** along these directions
- The data vary **minimally** orthogonally to these directions (ex: $\mathbf{u}_3$)

Mathematical formalization

Let $\mathcal{X}$ be our data, and $\{\mathbf{u}_i\}_{i\in[D]}$ a new orthonormal basis of $\mathbb{R}^D$.

- The **First Principal Component** $\mathbf{u}_1$ is chosen to maximize the variance of the projected data
- $\mathbf{u}_2$ is then chosen orthogonally to $\mathbf{u}_1$, and so on

Statistical model

For $\mathcal{X}$, we define:

- **Empirical mean:**

$$\overline{\mathcal{X}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i$$

- **Variance projected onto $\mathbf{u}_1$:**

$$v_{\mathbf{u}_1} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{u}_1^T\mathbf{x}_i - \mathbf{u}_1^T\overline{\mathcal{X}})^2 = \mathbf{u}_1^T\Sigma\mathbf{u}_1$$

where $\Sigma = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^T$ is the **covariance matrix**.

We therefore seek:

$$\mathbf{u}_1 = \arg\max_{\mathbf{u}} \mathbf{u}^T\Sigma\mathbf{u} \quad \text{subject to} \quad \mathbf{u}^T\mathbf{u} = 1$$

Solution by constrained optimization

We form the Lagrangian:

$$J(\mathbf{u}) = \mathbf{u}^T\Sigma\mathbf{u} + \lambda(1 - \mathbf{u}^T\mathbf{u})$$

The optimality conditions give:

$$\frac{\partial J(\mathbf{u})}{\partial \mathbf{u}} = 0 \quad \text{and} \quad \frac{\partial J(\mathbf{u})}{\partial \lambda} = 0$$

Which leads to:

$$\Sigma\mathbf{u}_1 = \lambda_1\mathbf{u}_1 \quad \Rightarrow \quad v_{\mathbf{u}_1} = \mathbf{u}_1^T\Sigma\mathbf{u}_1 = \lambda_1$$

Key interpretation: $(\mathbf{u}_1, \lambda_1)$ is an **eigenpair** of Σ.

Continuing the process, we obtain all principal components as the eigenpairs $\{(\mathbf{u}_i, \lambda_i)\}_{i\in[D]}$.

We can write compactly:

$$\Lambda = \mathbf{U}^\top \Sigma \mathbf{U}$$

with $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_D)$ and $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_D]$.

Variance-error duality

The projected variance is expressed as:

$$\nu_{\mathbf{u}_1} = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{u}_1^T (\mathbf{x}_i - \overline{\mathbf{x}}))^2$$

Maximizing this variance is equivalent to minimizing the quadratic approximation error:

$$\mathbf{u}_1 = \underset{\mathbf{u}^T\mathbf{u}=1}{\arg\min} \sum_{i=1}^{N} \|(\mathbf{x}_i - \overline{\mathbf{x}}) - [\mathbf{u}^T(\mathbf{x}_i - \overline{\mathbf{x}})]\mathbf{u}\|_2^2$$

Fundamental equivalence: Variance maximization $\Leftrightarrow$ Error minimization.

Physical interpretation

Consider a physical system $\mathcal{X}$ with unit masses at positions $\mathbf{x}_i$.

- **Inertia relative to a point a:**

$$\mathrm{I}_{\mathbf{a}}(\mathcal{X}) = \sum_{i=1}^{N} \mathrm{d}^2(\mathbf{a}, \mathbf{x}_i)$$

- **Huygens' theorem:** If $\mathbf{g}$ is the center of mass, then

$$\mathrm{I}_{\mathbf{a}}(\mathcal{X}) = \mathrm{d}^2(\mathbf{a}, \mathbf{g}) + \mathrm{I}_{\mathbf{g}}(\mathcal{X})$$

For a subspace $\mathcal{J}$ passing through $\mathbf{g}$:

$$\mathrm{I}_{\mathcal{J}}(\mathcal{X}) = \sum_{i=1}^{N} \mathrm{d}^2(\mathcal{J}, \mathbf{x}_i)$$

where $\mathrm{d}(\mathcal{J}, \mathbf{x}_i) = \|\mathbf{x}_i - \mathrm{Proj}_{\mathcal{J}}(\mathbf{x}_i)\|$.

Physical conclusion: A Principal Component is a **subspace of minimal inertia**.

Latent space structure

Fundamental properties

The basis $\{\mathbf{u}_1, \dots, \mathbf{u}_D\}$ of the latent space has:

- **Ordered eigenvalues:** $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_D$
- **Total variance:** $\sum_{d=1}^{D} \lambda_d = \mathrm{Tr}(\Sigma)$
- **Decorrelation:** $\mathbf{u}_d^\top \mathbf{u}_{d'} = 0$ (orthogonality)
- **Diagonal covariance matrix:** Λ is diagonal

Data transformation

The latent coordinates $\mathbf{y}_i$ are obtained by:

$$y_i(d) = \mathbf{u}_d^\top (\mathbf{x}_i - \overline{\mathbf{x}}) \quad \Rightarrow \quad \mathbf{y}_i = \mathbf{U}^\top (\mathbf{x}_i - \overline{\mathbf{x}})$$

This linear transformation combines three operations:

1. **Centering** on $\overline{\mathbf{x}}$
2. **Rotation** via $\mathbf{U}$
3. **Scaling** via $\Lambda^{1/2}$ (whitening)

Model structure

Underlying assumption

The latent space preserves the variance of the centered data, which assumes a model:

$$X_i \sim \mathcal{N}(\overline{\mathbf{x}}, \Sigma)$$

Important consequence: PCA will be less relevant for non-Gaussian data (ex: clustered data).

Exact transformation

At this stage, PCA is an **exact** and **invertible** transformation:

$$\mathbf{y}_i = \mathbf{U}^\top(\mathbf{x}_i - \overline{\mathbf{x}}) \quad \Leftrightarrow \quad \mathbf{x}_i = \overline{\mathbf{x}} + \mathbf{U}\mathbf{y}_i$$

The study of the **spectrum** $\{\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_D\}$ is crucial.

PCA for dimensional reduction

Low-rank approximation

According to the **Eckart-Young theorem**, for $K < D$:

- **Optimal approximation:** $\Sigma_K = \sum_{d=1}^{K} \lambda_d \mathbf{u}_d \mathbf{u}_d^\top$
- **Minimal error:** $\|\Sigma - \Sigma_K\|_F^2 = \sum_{d=K+1}^{D} \lambda_d$

By truncation:

$$\Sigma_K = \mathbf{U}_K \Lambda_K \mathbf{U}_K^\top$$

The transformed and reconstructed data become:

- **Reduced representation:** $\tilde{\mathbf{y}}_i = \mathbf{U}_K^\top \mathbf{x}_i \in \mathbb{R}^K$
- **Reconstruction:** $\tilde{\mathbf{x}}_i = \mathbf{U}_K \mathbf{U}_K^\top \mathbf{x}_i + \overline{\mathbf{x}}$

PCA geometry

Quality measures

- **Variance contribution:**

$$c_d = \frac{\lambda_d}{\sum_k \lambda_k}$$

- **Data alignment:**

$$\rho_d(\mathbf{x}_i) = \frac{(\mathbf{u}_d^\top \mathbf{x}_i)^2}{\|\mathbf{x}_i\|^2} = \cos^2(\angle(\mathbf{u}_d, \mathbf{x}_i))$$

Scale normalization

Each original dimension has its own scale (variance σ_d^2). PCA is more effective when all scales are comparable.

Solution: Use the **correlation matrix** instead of the covariance:

- **Scaling matrix:** $\mathbf{S} = \mathrm{diag}[\sigma_1^2, \dots, \sigma_D^2]$
- **Associated metric:** $\mathrm{d}_S^2(\mathbf{x}, \mathbf{y}) = (\mathbf{x} - \mathbf{y})^\top \mathbf{S}^{-1} (\mathbf{x} - \mathbf{y})$
- **PCA on normalized data:** Use Σ_S (correlation matrix)

Practical PCA: how-to guide

Step-by-step algorithm

1. **Compute** the empirical mean $\overline{\mathbf{x}}$
2. **Center** the data: $\mathbf{x}_i \leftarrow (\mathbf{x}_i - \overline{\mathbf{x}})$
3. **Form** the centered matrix $\mathbf{X}$
4. **Compute** $\Sigma = \frac{1}{N}\mathbf{X}\mathbf{X}^\top$
5. **Decompose** $\Sigma = \mathbf{U}\Lambda\mathbf{U}^\top$
6. **Select** K components
7. **Project** and/or **reconstruct** the data

Strategies for selecting K

- **Target dimension:** $K = d$ (desired dimension)
- **Preserved variance:** Choose K such that

$$\frac{\sum_{d=1}^{K} \lambda_d}{\sum_{d=1}^{D} \lambda_d} \geq \tau$$

  (ex: $\tau = 0.95$ for 95% variance)

- **Reconstruction error:** Choose K such that

$$\mathcal{L}(\mathbf{x}) = \sum_i \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2 \leq \epsilon$$

PCA and Linear Autoencoders

Linear Autoencoder architecture

- **Encoder:** $\mathbf{z}(\mathbf{x}_i) = W_{\text{enc}}\mathbf{x}_i$
- **Decoder:** $\tilde{\mathbf{x}}_i = W_{\text{dec}}\mathbf{z}(\mathbf{x}_i)$

Cost function

$$\theta^* = \arg\min_{W_{\text{enc}}, W_{\text{dec}}} \sum_{i=1}^{N} \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2$$

Equivalence with PCA

Theorem: A Linear Autoencoder trained on **centered** data performs exactly a PCA.

- **Encoder:** $W_{\text{enc}} = \mathbf{U}_K^\top$
- **Decoder:** $W_{\text{dec}} = \mathbf{U}_K$

Generalizations

- **Non-Linear Autoencoders:** Addition of hidden layers
- **Variational Autoencoders (VAE):** Modification of the cost function (ELBO)
- **Sparse Autoencoders:** Regularization constraints

Alternatives and extensions

Probabilistic PCA

Probabilistic reformulation with:

- **Latent distribution:** $\mathbb{P}(\mathbf{z}) = \mathcal{N}(0, \text{Id}_K)$
- **Conditional model:** $\mathbb{P}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(W\mathbf{z} + \mu, \sigma^2\text{Id}_D)$

Advantages:

- Explicit noise modeling
- Generative mode
- Estimation by Maximum Likelihood or EM

Kernel PCA

Non-linear extension via the **kernel trick**:

- **Implicit mapping:** $\Phi(\mathbf{x}_i)$ via kernel $k(\mathbf{x}_i, \mathbf{x}_j)$
- **PCA in feature space:** $\mathbf{k}(x_i, x_j) = \Phi(x_i)^\top \Phi(x_j)$

Local PCA

PCA applied locally on data neighborhoods to capture the **local intrinsic dimensionality**.

Computational limitations

The decomposition of Σ has complexity $O(D^3)$, which can be prohibitive for large dimensions.

Synthesis and conclusions

- **PCA** belongs to **linear** latent models
- It applies to **centered** data and uses **variance** as criterion
- It is an **exact** decomposition into **decorrelated** components
- It assumes a **normal** distribution of data
- It can be formulated as a **quadratic regression**
- It offers a **dimensional reduction** strategy based on low-rank approximation
- It can be used for **denoising** (Gaussian model)
- It is equivalent to a **Linear Autoencoder**
- It admits **stochastic** formulations and **non-linear** extensions

Deepening questions

- Explain how PCA uses variance as a criterion
- Show the equivalence between variance maximization and error minimization
- Explain the use of the Eckart-Young theorem in PCA
- Describe the practical steps to apply PCA to a dataset
- What information does PCA provide about the data?
- How to reconstruct data with $K < D$ components?
- What strategies to select the number of components to retain?
- Is PCA applicable to all types of data?
- Is it relevant to apply PCA to clustered data?
- Demonstrate the equivalence between PCA and Linear Autoencoder